

SYDNEY BOYS' HIGH SCHOOL

MOORE PARK, SURRY HILLS



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2001

MATHEMATICS

EXTENSION 1

Time allowed: 2 Hours
(plus five minutes reading time)
Examiner: E. Choy

DIRECTIONS TO CANDIDATES

- *ALL* questions may be attempted.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Start each question on a new answer sheet.
- Additional answer sheets may be obtained from the supervisor upon request.

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate examination paper for this subject.

Question 1. (12 marks)

- (a) Find the acute angle (correct to the nearest minute) between the lines $3x + 2y = 7$ and $4x - 3y = 2$. 2
- (b) Using the expansion of $\tan(\alpha - \beta)$, or otherwise, show that $\tan(-15^\circ) = \sqrt{3} - 2$. 2
- (c) Find $\lim_{x \rightarrow 0} \left(\frac{\sin 4x + \tan x}{x} \right)$. 2
- (d) Differentiate with respect to x : 2
- (i) $y = \ln(\cos x)$
- (ii) $y = \tan^{-1} 3x$
- (e) Solve $2\cos^2 x + 3\sin x - 3 = 0$, where $0 \leq x \leq 2\pi$. 2
- (f) Find the co-ordinates of the point P that divides the interval joining the points $A(-3,4)$ and $B(-1,0)$ externally in the ratio 4:3. 2

Question 2. (12 marks)

(a) Find the general solution of $\tan x = \sqrt{3}$. Give your answer in a concise, general form. 2

(b) How many different 9-letter “words” can be made from the letters of *ISOSCELES*? 2

(c) Find the domain and range of the function $y = \sin^{-1}(1 - \sqrt{x})$. 1

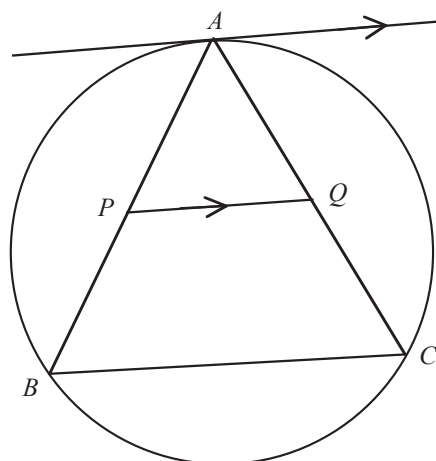
(d) Evaluate $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}}$. 2

(e) Find all solutions to $\frac{x}{x^2-1} > 0$. 2

(f) Given $AB = AC$, and that the tangent at A is parallel to PQ .

Prove:

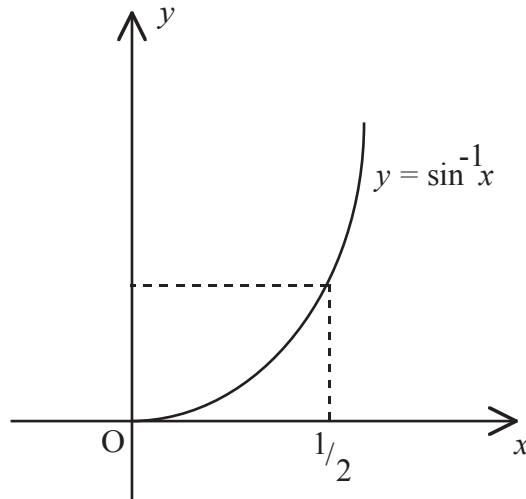
- (i) $AP=AQ$
- (ii) BC is parallel to the tangent at A .
- (iii) $PCBQ$ is a cyclic quadrilateral.



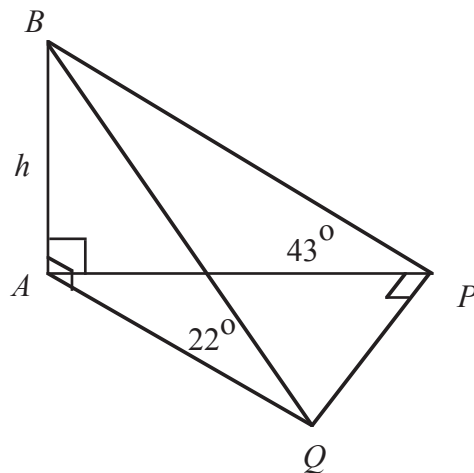
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Question 3. (12 marks)

- (a) Find the exact area bounded by the curve $y = \sin^{-1} x$, the x -axis, and the ordinate $x = \frac{1}{2}$ as shown in the diagram. 4



- (b) 4



The elevation of the top of a hill (B) from a place P due east of it is 43° , and from a place Q , due south of P , it is 22° . The distance from P to Q is 400m. If h is the height of the hill, show that

$$h^2 = \frac{160000}{\cot^2 22 - \cot^2 43}.$$

- (c) Find $\int \sec^2 x \cdot \tan^2 x \, dx$ using the substitution $u = \tan x$. 4

Question 4. (12 marks)

(a) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. **8**

(i) Find the co-ordinates of A , the point of intersection of the tangents to the parabola at P and Q .
(You may use the fact that equation of the tangent to the parabola $x^2 = 4ay$ at the point $T(2at, at^2)$ is $y = tx - at^2$.)

(ii) Suppose further that A lies on the line containing the focal chord which is perpendicular to the axis of the parabola.

(α) Show that $pq = 1$.

(β) Show that the chord PQ meets the axis of the parabola on the directrix.

(b) If $y = x^3 - 2x^2 + 3$ **4**

(i) find the equation of the tangent to the curve at $(2, 3)$, and

(ii) find the point at which the tangent meets the curve again.

Question 5. (12 marks)

- (a) Prove by mathematical induction that for positive integral n , $3^{3n} + 2^{n+2}$ is divisible by 5. **4**
- (b) By considering the function $f(x) = x^3 - 7$, use one step of Newton's method to find a better approximation to $\sqrt[3]{7}$ than 2. Leave your answer in exact fractional form. **3**
- (c) The speed v m/s of a point moving along the x -axis is given by $v^2 = 90 - 12x - 6x^2$, where x m is the displacement of the point from the origin. **3**
- (i) Prove that the motion is simple harmonic.
- (ii) Find the period, the centre of motion, and the amplitude.
- (d) (i) Prove that $\cos 2\theta = \frac{1-x^2}{1+x^2}$, where $x = \tan \theta$. **2**
- (ii) Use the above result to deduce that $\tan \frac{\pi}{8} = \sqrt{2} - 1$.

Question 6. (12 marks)

(a) Given $y = \sin^{-1}(\cos x)$: **4**

(i) Find $\frac{dy}{dx}$.

(ii) Evaluate $y = \sin^{-1}(\cos x)$ if $x = \pi$.

(iii) Sketch $y = \sin^{-1}(\cos x)$ for $-\pi \leq x \leq \pi$.

(b) Whilst playing tennis, Eric serves a ball from a height of 1.8 metres. If he hits the ball in a horizontal direction at a speed of 35 m/s, find (using $g = 10 \text{ ms}^{-2}$): **6**

(i) How long before the ball hits the ground.

(ii) How far the ball will travel before bouncing.

(iii) By how much the ball clears the net, which is 0.95 m high and 14 metres distant.

(c) (i) Find $\frac{d}{dx}(xe^x)$. **2**

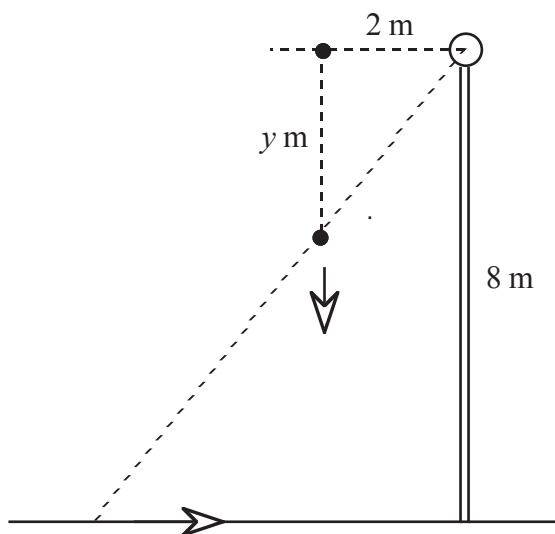
(ii) Use the result in Part (i) to evaluate $\int_0^1 xe^x dx$

Question 7. (12 marks)

- (a) A street lamp is 8 m high. A small object 2 m away from the lamp falls vertically downward.

6

- (i) Show that when the object has fallen y metres, the shadow it casts on the horizontal ground is $\frac{16}{y}$ metres from the base of the lamp.
- (ii) When the object has fallen 6 m, it is travelling at 10 m/s. At what speed is its shadow moving?
- (iii) At what height does the object have the same speed as its shadow?



- (b) A function $f(x)$ is defined by the rule $f(x) = (e^x - 1)\ln x$ for $0 < x \leq 1$.

6

- (i) Evaluate $f'(1)$.
- (ii) Using the fact that $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$, show that $f'(x) \rightarrow -\infty$ as $x \rightarrow 0$.
- (iii) Hence or otherwise show that $f(x)$ has a stationary value, and determine its nature.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



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Mathematics Extension 1

Sample Solutions

Q1 a) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

gradients $m_1 = -\frac{1}{2}$, $m_2 = \frac{4}{3}$

(2) $\tan \theta = \frac{-\frac{1}{2} - \frac{4}{3}}{1 + (-\frac{1}{2}) \cdot \frac{4}{3}}$
 $= \frac{11}{2}$

$\theta = 70.34^\circ$

(e) $2(1 - \sin x) + 3 \sin x - 3 = 0$

$2 - 2 \sin x + 3 \sin x - 3 = 0$

$2 \sin x - 3 \sin x + 1 = 0$

$(2 \sin x - 1)(\sin x - 1) = 0$

$\sin x = \frac{1}{2}$ $\sin x = 1$

$x = \frac{\pi}{6}, \frac{5\pi}{6} \text{ or } \frac{\pi}{2}$

(b) $\tan(30^\circ - 45^\circ) = \frac{\tan 30^\circ - \tan 45^\circ}{1 + \tan 30^\circ \tan 45^\circ}$

(2) $= \frac{\frac{1}{\sqrt{3}} - 1}{1 + \frac{1}{\sqrt{3}} \cdot 1} = \frac{\sqrt{3} - 3}{3 + \sqrt{3}}$
 $= \frac{6\sqrt{3} - 12}{6} = \sqrt{3} - 2$

(f) $k = 4$, $l = 3$

(5, 12) (2)

(c) $\lim_{x \rightarrow 0} \frac{\sin 4x}{x} + \lim_{x \rightarrow 0} \frac{\tan x}{x}$

$\neq \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} + \lim_{x \rightarrow 0} \frac{\tan x}{x}$

(2) $= 5$

(d) (i) $y = \ln(\cos x)$

$\frac{dy}{dx} = \frac{1}{\cos x} \cdot (-\sin x)$

(1) $= -\tan x$

(ii) $y = \tan^{-1} 3x$

(1) $\frac{dy}{dx} = \frac{3}{1 + 9x^2}$

Question 2

(a) $\tan x = \sqrt{3}$

$$\therefore x = 180n + \tan^{-1}(\sqrt{3})$$

$$\boxed{x = 180n + 60} \quad \text{or} \quad \boxed{x = n\pi + \frac{\pi}{3}}$$

b) $\frac{9!}{2!3!} = \begin{cases} \text{repetition of } S \text{ (x3)} \\ \text{repetition of } E \text{ (x2)} \end{cases}$

$$\boxed{30240}$$

c) domain: $-1 \leq 1 - \sqrt{x} \leq 1$

$$\therefore -2 \leq -\sqrt{x} \leq 0$$

$$\therefore 0 \leq \sqrt{x} \leq 2$$

$$\therefore \boxed{0 \leq x \leq 4} \quad \left(\frac{1}{2}\right)$$

range: $-\frac{\pi}{2} \leq \sin^{-1}u \leq \frac{\pi}{2}$

$$\therefore \boxed{-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}} \quad \left(\frac{1}{2}\right)$$

d) $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}} = \sin^{-1}\left(\frac{x}{\sqrt{3}}\right) \Big|_0^{\sqrt{3}}$

$$= \sin^{-1}(1) - \sin^{-1}(0)$$
$$= \frac{\pi}{2}$$

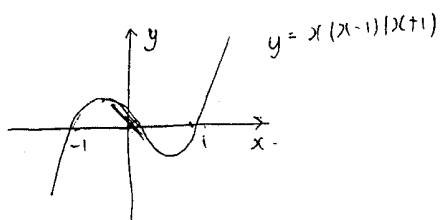
$$(e) \quad \frac{x}{x^2-1} > 0 \quad \boxed{x \neq \pm 1}$$

$$\therefore \frac{x}{(x+1)(x-1)} > 0$$

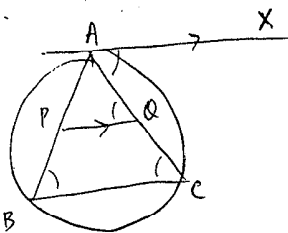
$$[x(x+1)^2(x-1)^2]$$

$$\therefore (x+1)(x-1)x > 0$$

$$\therefore \boxed{-1 < x < 0, x > 1}$$

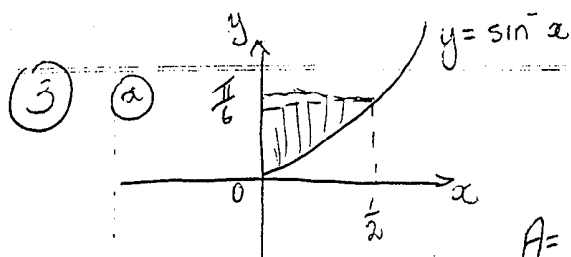


(f) (i) $\because AB = AC$
 $\therefore \angle ABC = \angle ACB$ (base angles of isos. Δ)
 $\hat{x}AC = \hat{A}BC$ (alternate segment theorem)
 $\hat{x}AC = \hat{AQP}$ (alternate angles)
 $\therefore \hat{AQP} = \hat{ACB}$
 $\therefore PQ \parallel BC$
 (corresponding angles equal)
 $\therefore \hat{APQ} = \hat{AQP}$
 $\therefore AP = AQ$ (isos. Δ)



(ii) $BC \parallel PQ$ & $PQ \parallel AX$
 $\therefore BC \parallel AX$

(iii) $\hat{APQ} = \hat{ACB}$ (from (i))
 $\therefore$ exterior angle equals opposite interior angle
 $\therefore PQCB$ is cyclic quad.



$$\sin^{-1} \frac{1}{2} = 30^\circ = \frac{\pi}{6}$$

If $y = \sin^{-1} x$ then $\sin y = x$

$$A = \int_0^{\frac{\pi}{6}} -\sin y \, dy$$

$$= -\cos y \Big|_0^{\frac{\pi}{6}}$$

$$= -\cos \frac{\pi}{6} + \cos 0$$

$$= -\frac{\sqrt{3}}{2} + 1$$

$$= 1 - \frac{\sqrt{3}}{2}$$

Now area rectangle is $\frac{1}{2} \times \frac{\pi}{6} = \frac{\pi}{12}$ (exact)

$$\text{area required is } \frac{\pi}{12} - \left(1 - \frac{\sqrt{3}}{2}\right) = \frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2} \quad u^2$$

(b) In $\triangle BAQ$, $\tan 22^\circ = \frac{h}{AQ}$ (4)
 $h = AQ \tan 22^\circ$
 $AQ = \frac{h}{\tan 22^\circ} = h \cot 22^\circ$

In $\triangle BAP$ $\tan 43^\circ = \frac{h}{AP}$
 $h = AP \tan 43^\circ$
 $AP = \frac{h}{\tan 43^\circ} = h \cot 43^\circ$

Now In $\triangle APQ$, $AP^2 + PQ^2 = AQ^2$ (4)

$$AP^2 - AQ^2 = -160000$$

$$h^2 \cot^2 43^\circ - h^2 \cot^2 22^\circ = -160000$$

$$h^2 = \frac{-160000}{\cot^2 43^\circ - \cot^2 22^\circ} = \frac{160000}{\cot^2 22^\circ - \cot^2 43^\circ}$$

$$3 \text{ (c)} \int \sec^2 x \tan^2 x \, dx$$

$$= \int u^2 \cdot du$$

$$= \frac{u^3}{3} + C$$

$$= \frac{\tan^3 x}{3} + C$$

$$\begin{aligned} \text{let } u &= \tan x \\ \frac{du}{dx} &= \sec^2 x \\ du &= \sec^2 x \cdot dx \end{aligned}$$

(4)

Q4 (a) (i) $y = px - ap^r$ — (1)

$y = qx - aq^r$ — (2)

$0 = (p-q)x - a(p^r - q^r)$

$x = \frac{a(p^r - q^r)}{p-q}$

$x = a(p+q)$

$y = pa(p+q) - ap^r$

$= ap^r + apq - ap^r$

$y = apq$

$\therefore A \text{ is } (a(p+q), apq)$ 3

(ii) If $y = a$ then $apq = a$
 $pq = 1$ 1

(iii) Check $y = \frac{1}{2}(p+q)x - apq$

show $\frac{y - ap^r}{x - ap^r} = \frac{aq^r - ap^r}{2aq - 2ap} = \frac{q+p}{2}$

$y - ap^r = \left(\frac{p+q}{2}\right)x - 2ap(p+q)$

$y - ap^r = p+q x - ap^r - apq$

$y = \frac{p+q}{2}x - apq$ 2

now if $x = 0$ $y = -apq$
 $y = -a$ because $pq = 1$ 2

-8-8+3

(b) $y = x^3 - 2x^2 + 3$
 $\frac{dy}{dx} = 3x^2 - 4x \downarrow \downarrow$
 $m = 3x + 4x^2 \downarrow 1$
 $m = 4$

$\frac{y-3}{x-2} = 4$
 $y-3 = 4x-8$
 $y = 4x-5$
2

Solve $x^3 - 2x^2 + 3 = 4x - 5$
 $x^3 - 2x^2 - 4x + 8 = 0 \downarrow \downarrow$
restrain 2, 2, 2
 $2 + 2 + 2 = 2$
 $2 = -2$
 $\therefore (-2, -13)$ 2

5(a) Test for $n=1$, $S_1 = 3^3 + 2^3$

$$= 27 + 8$$

$= 35$ which is divisible by 5. ✓

Assume true for $n=k$, i.e. $S_k = 3^{3k} + 2^{k+2}$ ✓
 $= 5P$ where $P \in \mathbb{Z}$.

Now test for $n=k+1$, i.e. $S_{k+1} = 3^{3k+3} + 2^{k+3}$
 $= 5Q$ where $Q \in \mathbb{Z}$.

$$S_{k+1} = 27(5P - 2^{k+2}) + 2 \cdot 2^{k+2}$$

$$= 5 \cdot 27P - (27-2)2^{k+2}$$

$$= 5 \{ 27P - 5 \cdot 2^{k+2} \}$$

$$= 5Q$$

∴ True for $n=k+1$ if true for $n=k$.

Now true for $n=1$ so true for $n=2$ and so on for all integer n .

(b) $f(x) = x^3 - 7$, $f'(x) = 3x^2$ ✓

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 2 - \frac{8-7}{3 \cdot 4}$$

$$= \frac{23}{12}$$

(c) $\frac{v^2}{2} = 45 - 6x - 3x^2$

$$\ddot{x} = \frac{d}{dx} \left(\frac{v^2}{2} \right)$$

$$= -6 - 6x$$

$$= -6(x+1)$$

$$= -(\sqrt{6})^2 X \text{ where } X = x+1$$

∴ Motion is SHM with centre of motion $= -1$.

$$m = \sqrt{6} \text{ so period} = \frac{2\pi}{\sqrt{6}} = \frac{\sqrt{6}\pi}{3}$$

$$v^2 = -6(x^2 + 2x + 1) + 90 + 6$$

$$= 96 - 6(x+1)^2$$

$$= 6 \{ 4^2 - (x+1)^2 \}$$

Amplitude $= 4$

$$5(d)(i) \text{ RHS} = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \times \frac{\cos^2 \theta}{\cos^2 \theta}$$

$$= \frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta + \cos^2 \theta}$$

$$= \cos 2\theta \quad \checkmark$$

$$= \text{L.H.S.}$$

$$(ii) \text{ If } \theta = \pi/8, \cos 2\theta = \frac{1}{\sqrt{2}} = \frac{1 - x^2}{1 + x^2}$$

$$1 + x^2 = \sqrt{2} - \sqrt{2} x^2$$

$$x^2(1 + \sqrt{2}) = \sqrt{2} - 1$$

$$x^2 = \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$$

$$= \frac{(\sqrt{2} - 1)^2}{2 - 1} \quad \checkmark$$

$$x = \sqrt{2} - 1 \text{ as } \tan \pi/8 \text{ is in 1st quadrant.}$$

(a) Question 6

(i) $y = \sin^{-1}(\cos x)$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\cos^2 x}} \cdot -\sin x$$

$$= \frac{-\sin x}{\sqrt{\sin^2 x}}$$

2

$$= \frac{-\sin x}{|\sin x|}$$

$$= -1 \text{ for } 0 < x < \pi$$

$$= 1 \text{ for } -\pi < x < 0$$

(ii) $y = \sin^{-1}[\cos \pi]$

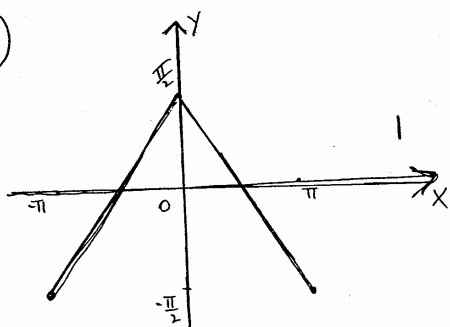
$$= \sin^{-1}[-1]$$

1

$$= -\sin^{-1}[1]$$

$$= -\frac{\pi}{2}$$

(iii)



(b) $\dot{x} = V \cos \theta = 35 \cos 0 = 35$

$$\dot{y} = 35 \sin 0 - 10t = -10t$$

$$y = 1.8 - 5t^2$$

$$x = Vt \cos \theta = 35t$$

(i) Strikes ground when $y=0$

$$\therefore 0 = 1.8 - 5t^2$$

$$t = 3/5 \text{ sec.}$$

✓

(ii) $x = 35 \times \frac{3}{5} = 21 \text{ m}$ 2

(iii) When $x=14$, $14 = 35t$

$$\therefore t = \frac{2}{5}$$

$$\text{When } t = \frac{2}{5}, y = 1.8 - 5\left(\frac{2}{5}\right)^2$$

2

$$\therefore y = 1 \text{ m}$$

$$\therefore \text{Clears net by } 1 - 0.95 \text{ m} = 5 \text{ cm.}$$

(c)

(i) $\frac{d}{dx}(xe^x) = xe^x + 1 \cdot e^x = e^x(x+1)$

(ii) $\frac{d}{dx}(xe^x - e^x) = xe^x$

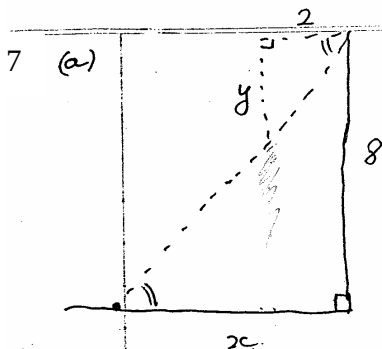
$$\therefore \frac{d}{dx}[xe^x - e^x] = xe^x \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} 1$$

$$\Rightarrow \int xe^x dx = xe^x - e^x \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} 1$$

$$\therefore \int_0^1 xe^x dx = [xe^x - e^x]_0^1 \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} 1/2$$

$$= [e - 0] - [0 - e^0]$$

$$= e - 1$$



(i) Triangles are similar

$$\therefore \frac{x}{2} = \frac{8}{y}$$

$$x = \frac{16}{y}$$

(ii) $\frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$

$$= -\frac{16}{y^2} \cdot 10$$

When $y = 6$, $\frac{dx}{dt} = -\frac{16}{36} \cdot 10$
 $= -\frac{40}{9} \text{ m/s.}$

(iii) $\frac{dy}{dt} = \frac{dx}{dt}$

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{dx}{dt}$$

$$\frac{dy}{dx} = 1$$

$$-\frac{16}{y^2} = -1 \text{ (speed)}$$

$$y = 4$$

(b)

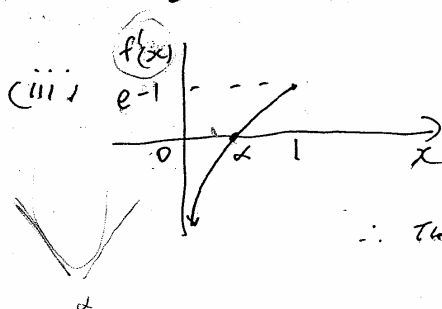
(i) $f(x) = (e^x - 1) \ln x$

$$f'(x) = \frac{e^x - 1}{x} + e^x \ln x$$

$$f'(1) = e - 1 + 0 = e - 1$$

(ii) As $x \rightarrow 0$, $f'(x) \rightarrow -\infty$

since $\ln x \rightarrow -\infty$ as $x \rightarrow 0$



$$f'(x) = 0 \text{ for } 0 < x < 1$$

Let this root be α .

For $x < \alpha$, $f'(x) < 0$

$x > \alpha$, $f'(x) > 0$

$\therefore$ The stationary point is a local minimum.